WORKED SOLUTION

STEP I 1995, Question 11

Mechanics • Friction

STEP I 1995, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q11

The key idea

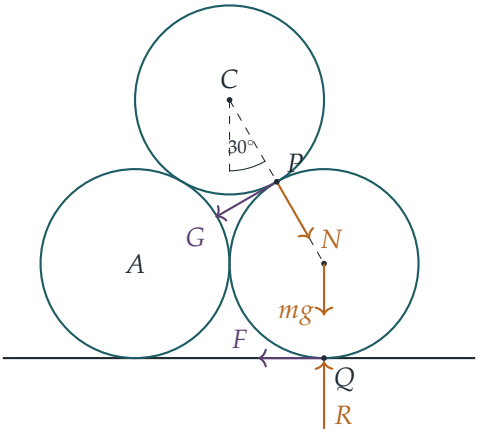
Look at one of the bottom cylinders. Its weight and the normal reaction from the top cylinder both pass through its centre, so taking moments about the centre involves only the two friction forces, and they must be equal in size. One horizontal resolution then fixes the ratio of friction to normal reaction at the contact between the cylinders, and the geometry of three touching circles (angles of 30° and 60°) does the rest.

Solution

The model

We treat the cylinders as uniform rigid cylinders of radius r and work in a cross-section perpendicular to the axes, so each cylinder is a circle with its weight acting at the centre. Every contact has the same coefficient of friction μ. Call the bottom cylinders A (left) and B (right), and the top one C. The centres of A and B are $2r$ apart at height r; the centre of C is $2r$ from each, so the triangle of centres is equilateral and the line from the centre of C to the centre of B makes 30° with the vertical.

For now we assume A and B exert no force on each other; we come back to this at the end. By symmetry it is enough to look at B and at the right-hand contact of C.



The forces on B are:

- its weight mg at its centre;
- at Q, the ground contact: a normal reaction R upwards and a friction force F horizontally towards A;
- at P, the contact with C: a normal reaction N along the line of centres (down and outwards, at 30° to the vertical), and a friction force G along the common tangent, which we take as acting down and towards A, that is at 30° below the horizontal.

If either friction force really acts the other way, its value will simply come out negative.

Moments about the centre of B

The weight passes through the centre of B, and so does N, since a normal reaction between two circles acts along the line of their centres. Both friction forces act along tangents, at perpendicular distance r from the centre. F, at the bottom and pointing towards A, turns B clockwise in the figure; G, at P and pointing down the tangent towards A, turns it anticlockwise. Hence

$$Fr = Gr, \qquad \text{so} \qquad F = G.$$

Resolving horizontally for B

N has an outward horizontal component $N\sin 30° = \frac{1}{2}N$. Both F and the horizontal component $G\cos 30° = \frac{\sqrt{3}}{2}G$ of G point inwards. So

$$\tfrac{1}{2}N = F + \tfrac{\sqrt{3}}{2}G = G\left(1 + \tfrac{\sqrt{3}}{2}\right),$$

and therefore

$$\frac{G}{N} = \frac{1}{2+\sqrt{3}} = \frac{2-\sqrt{3}}{(2+\sqrt{3})(2-\sqrt{3})} = 2-\sqrt{3}.$$

This is positive, so G does act in the direction we assumed, and so does $F = G$.

For the contact at P to hold without slipping we need $G \leq \mu N$, that is

$$\mu \geq 2 - \sqrt{3}.$$

The contact with the ground

We must also check that the ground contact does not demand more friction. For C, the forces from B are N up and to the left at 30° to the vertical, and G up the tangent at 30° to the horizontal. Their vertical components are $\frac{\sqrt{3}}{2}N$ and $\frac{1}{2}G$, and by symmetry A contributes the same. Resolving vertically for C,

$$2\left(\tfrac{\sqrt{3}}{2}N + \tfrac{1}{2}G\right) = mg \implies N\left(\sqrt{3} + 2 - \sqrt{3}\right) = mg \implies N = \tfrac{1}{2}mg.$$

Resolving vertically for the whole system, which has weight $3mg$ shared equally by symmetry, $R = \frac{3}{2}mg$. Then

$$\frac{F}{R} = \frac{(2-\sqrt{3}) \cdot \frac{1}{2}mg}{\frac{3}{2}mg} = \frac{2-\sqrt{3}}{3} < 2 - \sqrt{3},$$

so whenever $\mu \geq 2 - \sqrt{3}$ the ground contact holds as well. The contact between the cylinders is the critical one.

> **Ask yourself** A and B are touching. Why may we ignore any force between them?

They can only push each other apart, so suppose A pushes B outwards with a force $S \geq 0$ along the line of centres. This force passes through the centre of B, so the moment equation still gives $F = G$, but the horizontal resolution becomes $\frac{1}{2}N + S = G\left(1 + \frac{\sqrt{3}}{2}\right)$. Meanwhile the vertical equation for C still holds, and solving the two together shows that G/N increases as S increases (at $S = 0$ it is $2 - \sqrt{3}$). The problem is statically indeterminate, since S is not fixed by the equations, and the least friction is needed when $S = 0$. So the minimum possible coefficient of friction is $2 - \sqrt{3}$, as required.

Checks

- $2 - \sqrt{3} = \tan 15^\circ \approx 0.268$. So at the cylinder contact the total reaction makes exactly 15° with the normal, half the 30° angle in the geometry. That is a pleasing sign the algebra is right.
- The ground ratio $(2 - \sqrt{3})/3 \approx 0.089$ is a third of the cylinder ratio, which makes sense: the ground carries $\frac{3}{2}mg$, three times N, but the same size of friction.
- A numerical check, substituting into all the force and moment equations for B and C and minimising over the unknown push $S \geq 0$, gave a least μ of $0.267949\ldots$, agreeing with $2 - \sqrt{3}$.

Common slips

- Taking moments about the ground contact or about P instead of the centre. This works, but it brings in N and gives messier algebra; the centre removes both mg and N at once.
- Checking only the ground. The ground contact needs much less friction, so it gives the wrong critical value.
- Forgetting that the friction on C acts *up* the tangent, by Newton's third law, when resolving vertically for C.

Beyond STEP

The force S between the bottom cylinders cannot be found from the equilibrium equations: the structure is *statically indeterminate*, and the question sensibly asks for the minimum friction over all possible values of S. In engineering such systems are settled by modelling elastic deformation, and the same issues come up in the mechanics of granular materials such as stacked pipes or piles of sand. For a careful treatment of statics, friction and rigid-body equilibrium, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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